

Manipal Hackathon 2026



Team Name: DataBaes

Track: Sustainable Cities and Communities (SDG 11)

Problem Statement: Cascading Failure - When One Failure Becomes Many

Solution Title: Ripple

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Solution

THE PROBLEM, AND OUR APPROACH

A single flooded bridge can cut off a hospital nobody thought was at risk.

R

Real, not hypothetical

Built on real OpenStreetMap positions and NASA / USGS SRTM elevation data.

=

Equity-weighted impact

Every failure is scored by who it affects, not just how many assets fail.

%

Capacity-aware, not binary

A backup tie can rescue a failed pump, but only up to its rated capacity.

WHAT THE REAL DATA REVEALED



Every hospital sits at 24 to 102 metres. The real risk is two low crossings the elevation data itself surfaced.

Malpe Bridge: 0.0m. A creek crossing near Kaup: 4.0m. Cut either one and people lose hospital access, with no building ever flooding.

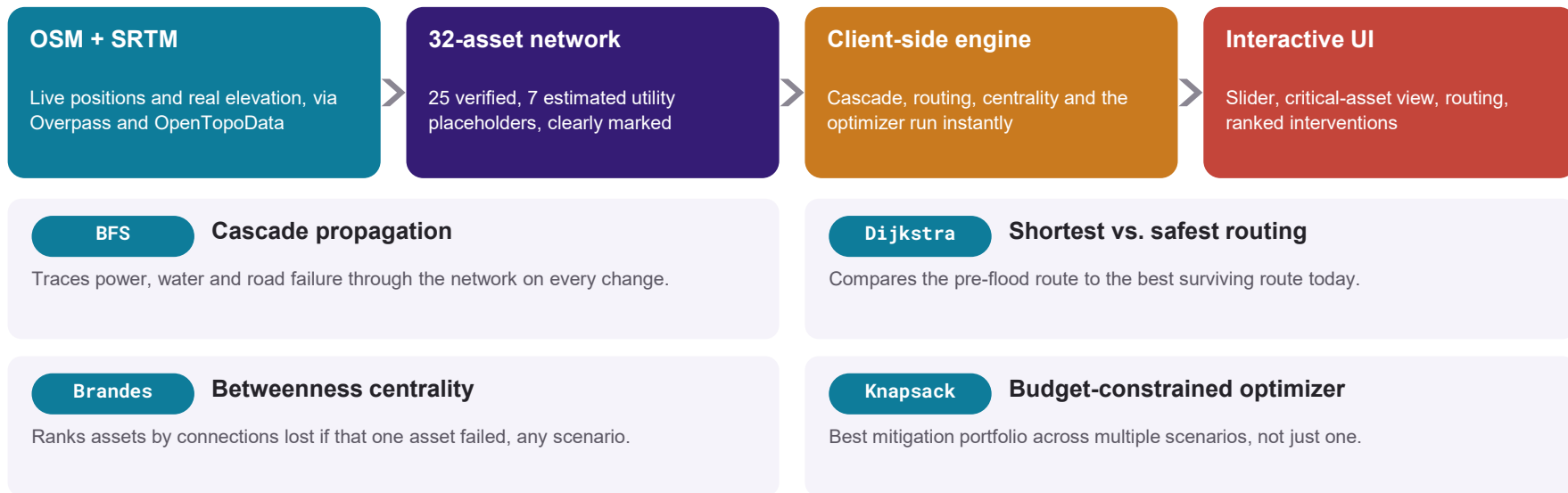
The elevation data revealed this risk. We did not design it in.



Technical Implementation

ARCHITECTURE, ALGORITHMS AND WHAT IS NEW

Real data in, an interactive simulation out, entirely in the browser



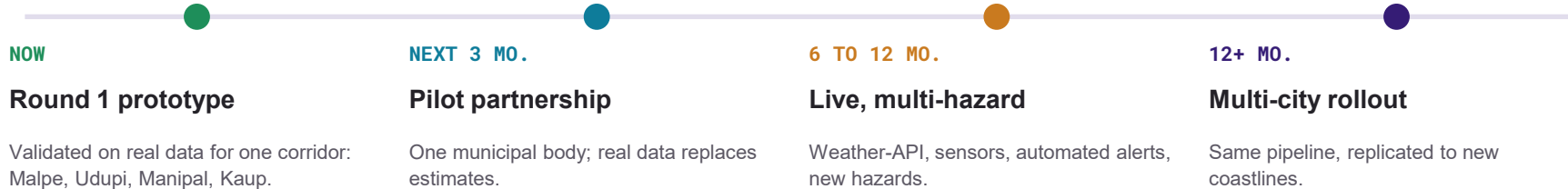
The overload model, capacity-limited backup ties between substations and pumps, was built specifically for this brief.



Feasibility

PRACTICALITY, ROADMAP AND INCLUSIVITY

A working prototype today, a path to real deployment



INCLUSIVITY AND REAL-WORLD FIT



Equity by design

The impact score itself weights vulnerable localities higher, by construction, not as an add-on.



Priced for who needs it

Slider-based, usable by field disaster-response staff, not a specialist GIS tool few can afford.



Honest about confidence

Verified positions are marked separately from estimates, useful even where clean data does not exist.



Business Strategy

MARKETING, MONETISATION AND SUSTAINABILITY

Sell to who already owns this problem, on three revenue lines

TARGET AUDIENCE

Primary: district and state disaster-management authorities.

Secondary: urban planning consultancies.

Tertiary: infrastructure insurers pricing flood risk.

OUTREACH PLAN

Pilot with Udupi's own district disaster-management authority, our own map's geography.

Open-source the engine for developer adoption.

Presence at Smart Cities Mission and AMRUT 2.0 forums.

1

B2G SaaS licensing

Per-city annual licence, tiered by assets modeled.

2

Data onboarding and consulting

Paid engagement to ingest a city's real dependency data.

3

Insurer data licensing

Longer term: flood-risk scoring as a data product.

The open-core engine stays free for adoption and research credibility; onboarding a new city is largely configuration, not new development, so three independent revenue lines reduce dependence on any one source.